

TRUSTWORTHY ADAPTATION WITH FEW-SHOT LEARNING FOR HAND GESTURE RECOGNITION

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ABSTRACT

This work is motivated by potentials of Deep Neural Networks (DNNs)-based solutions in improving myoelectric control for trustworthy Human-Machine Interfacing (HMI). In this context, we propose the Trustworthy Few Shot-Hand Gesture Recognition (TFS-HGR) framework as a novel DNN-based architecture for performing Hand Gesture Recognition (HGR) via multi-channel surface Electromyography (sEMG) signals. The main objective of the TFS-HGR framework is to employ Few-Shot Learning (FSL) formulation with a focus on transferring information and knowledge between source and target domains (despite their inherent differences) to address limited availability of training data. The NinaPro DB5 dataset is used for evaluation purposes. The proposed TFS-HGR achieves a performance of 83.17% for new repetitions with few-shot observations, i.e., 5-way 10-shot classification. Moreover, the TFS-HGR with the accuracy of 75.29% also generalize to new gestures with few-shot observations, i.e., 5-way 10-shot classification.

Index Terms— Few-Shot Learning (FSL), surface Electromyographic (sEMG), Hand Gesture Recognition (HGR), Attention Mechanism, Temporal Convolution

1. INTRODUCTION

Recent developments in rehabilitation and assistive technologies combined with advances in Machine Learning (ML) and Deep Neural Networks (DNNs) have led to the development of improved myoelectric prosthesis control systems. The surface electromyographic (sEMG) signals are especially used to classify limb movements and improve the quality of myoelectric prosthesis control systems, which enhance living conditions of hand amputated individuals. The sEMG signals are collected by wrapping sensors around the skin to measure the muscle fibers' action potentials. Because various applications of hand gesture recognition, such as prosthetic control, involve continuous and long-term use, it is essential to develop a learning algorithm that incorporates prior knowledge and experience gathered from the source domain with new information. In addition, there is a need to develop adaptive learn-

ing algorithms with the aim of designing a classifier, which can be used for new gestures based on only a few observations through the fast learning approach. This is a challenging task since many factors such as electrode shifts and muscle fiber lengthening/shortening can affect the gathered sEMG signals. Moreover, the neurophysiology differences between users result in more inconsistencies between different conditions [1, 2]. In this context, by taking advantage of Few-Shot Learning (FSL) the gathered experience from several source can speed up the recognition process for a new gesture. Thus, the learning process does not start from the beginning every time for an unseen gesture. In this regard, the paper focuses on hand gesture classification [3–13] using FSL to reduce the critical challenge of variability in the characteristics of sEMG signals. In other words, we propose a framework known as Trustworthy Few Shot-Hand Gesture Recognition (TFS-HGR) that utilizes a combination of temporal convolutions and attention mechanisms. The proposed TFS-HGR framework allows a myoelectric controller that has built based on background data to adapt the variations in stochastic characteristics of sEMG signals. The adaptation can be achieved with a small number of new observations making it suitable for clinical implementations and practical applications.

The paper is organized as follows: Section 1.1 provides an overview of relevant literature. In Section 2, we describe the proposed TFS-HGR architecture. Experimental results and different evaluation scenarios with dataset used in development of the proposed architecture are presented in Section 3. Finally, Section 4 concludes the paper.

1.1. Literature Review

A common strategy used to identify hand movements with DNN-based models is to consider 2/3 of the gesture trials of each user for the training set, while the rest of trials constitute the test set [5–13]. Despite extensive advances in myoelectric prostheses in recent DNN-based models, their performance is significantly reduced when data are limited. Collecting large data sets for training may be possible in research laboratories, but this is not a practical method for real applications. Therefore, the use of domain adaptation algorithms is a fundamen-

tal step in bridging the gap between real-world practice and academic achievement in the laboratory. In this regard, some researches such as [14, 15] utilize Transfer Learning (TL)-based algorithms to take advantage the knowledge obtained from source users and accelerate the learning process for new target users. In addition, Reference [16] adopts a TL-based algorithm and provides an adaptive calibration of electrode array shifts which enhances the robustness of the myoelectric control system. Moreover, in References [17, 18], the authors use a long-term recording of sEMG data to train and test various DNN-based architectures to provide a robust algorithm against non-stationary nature of sEMG signals varying over days.

Most of the existing techniques (such as [17, 18]) have been tested on data of limited dimension, with respect to the number of users and/or the number of hand gestures. Therefore, their generalizability to a larger population with an increased number of gestures has not been evaluated. Therefore, It is not clear how these models would perform on scenarios with a larger number of users and gestures. Therefore, despite the recent targeted focus on improving robustness in DNN-based myoelectric control models, more research is still needed, particularly on the design of new classification models, a unified framework for FSL, and domain adaptation algorithms. The paper, inspired by [19], makes the following significant contributions to the growing research in this area:

- by proposing a FSL framework that allows the myoelectric controller that has been built on background data from source gestures/repetitions, to adapt to the changes in the EMG signal characteristics from a target gesture/repetition with minimal efforts and a few observations, making it suitable for clinical settings and practical applications.
- By proposing the TFS-HGR framework, which utilizes a combination of temporal convolutions and attention mechanisms, we provide a novel venue for adopting few-shot learning, to not only reduce the training time, but also to eventually mitigate the significant challenge of variability in the characteristics of sEMG signals.

2. THE PROPOSED TFS-HGR ARCHITECTURE

Fast learning is an indication of human intelligence, which involves recognizing an object just by looking at a few examples or quickly repeating an action several times. There is a need to develop adaptive learning methods focusing on designing classifications for myoelectric control systems, which can be employed for new gestures based on only a few observations through a fast learning approach. In this regard, by inspiring from Reference [19], we propose the FSL-based architecture which utilized the temporal convolutions along with attention mechanism. In the following, first, we briefly

describe the FSL formulation, and then provide more details about the building blocks of TFS-HGR architecture.

2.1. The FSL Formulation

Within FSL formulation, we are dealing with tasks where each task $\mathcal{T}_j$ corresponds to a dataset $\mathcal{D}_j$ that splits into two parts, a support-set $\mathcal{D}_j^{support}$, and a query-set $\mathcal{D}_j^{query}$. The $\mathcal{D}_j^{support}$ consists of a sequence of observation-label pairs, while the $\mathcal{D}_j^{query}$ consists of an unlabelled observation. Therefore, based on the labels in the $\mathcal{D}_j^{support}$, the network is going to predict the missing label in the $\mathcal{D}_j^{query}$. The parameters of the network are optimized using the loss between the prediction and the ground truth label in the $\mathcal{D}_j^{support}$ over mini-batches of tasks. During the test procedure, unseen tasks are sampled from a different task distribution (i.e., $\tilde{\mathcal{T}}_j \sim p(\tilde{\mathcal{T}})$) that is similar in nature to $p(\mathcal{T})$.

More specifically, we follow reference [19] and define the N -way k -shot classification task as such that the support-set consists of k examples for each of N unique classes. The N classes are randomly selected from the overall dataset. Each observation in the support-set is concatenated with its corresponding label constituting the $N \times k$ time-step for the network. These $N \times k$ observation-label pairs are followed by an unlabelled observation which is sampled from one of the N classes. This observation is concatenated with a null label. The network is supposed to predict the missing label of the last $(N \times k + 1)^{th}$ time-step.

2.2. The Building Blocks of TFS-HGR Architecture

The proposed TFS-HGR architecture consists of several structural blocks whose algorithms are provided in Algorithm 1, Algorithm 2, and Algorithm 3. Each block receives an input with size $(C_{in} \times l)$, where C_{in} represents the input dimensionality and l is the length of sequence which is equal to $(N \times k + 1)$. It should be noted that before feeding the input to the proposed TFS-HGR framework, it becomes a feature vector. For this target, we used an *Embedding Module*.

More specifically, For performing HGR, it is common to segment sEMG data through a window before feeding it as input to neural networks. Therefore, in this work, before feeding the sEMG data, it is segmented by a window with size $W \times N$, where W shows the length of the window and N indicates the number of sensors. Then, an Embedding Module is used to convert the input with size $W \times N$ to a feature vector. This feature vector is used as input for subsequent modules in the proposed TFS-HGR network. In this paper, we adopted 2 Embedding Module. The first one, is using the Fully Connected (FC) layers for extracting the feature vector. The second case is the use of Long Short Term Memory (LSTM) as the Embedding Module for extracting this feature vector.

The next building block is *The TemporalBlock Module* which consists of two Dilated Causal 1D-Convolutions with dilation factor d , kernel size k_s , and f filters. Each Dilated Causal 1D-Convolution is followed by ReLU activation func-

Table 1: Experiment 1: 5-way 1-shot, 5-way 5-shot, and 5-way 10-shot classification accuracies based on *new repetitions with few-shot observation*. In this experiment, we adopted two different Embedding Modules: (i) FC Embedding and (ii) LSTM Embedding

The Embedding Module	5-way Accuracy (%) $\pm$ STD		
	1-shot	5-shot	10-shot
FC Embedding	66.75 $\pm$ 0.14	79.48 $\pm$ 0.13	79.56 $\pm$ 0.15
LSTM Embedding	69.61 $\pm$ 0.15	78.75 $\pm$ 0.14	83.17 $\pm$ 0.15

Algorithm 1 THE TEMPORALBLOCK MODULE

function: TemporalBlock(input, dilation factor d , kernel size k_S , number of filters f):

- 1: output1 = CausalConv(input, d , k_S , f)
- 2: activation1 = relu(output1)
- 3: output2 = CausalConv(activation1, d , k_S , f)
- 4: activation2 = relu(output2)

return concat(input, activation2)

Algorithm 2 THE TEMPORALCONVNET MODULE

function: TemporalConvNet(input, sequence length $l = (N \times k + 1)$, kernel size k_S , number of filters f):

- 1: $Z = \lceil \log_2 l \rceil$
- 2: **for** i in $0, \dots, Z - 1$ **do**
- 3: input = TemporalBlock(input, 2^i , k_S , f)
- 4: **end for**

return input

tion. Finally, the input and output are concatenated (Algorithm 1).

The *TemporalConvNet Module* consists of $Z = \lceil \log_2 l \rceil$ TemporalBlock Modules whose dilation factor increases exponentially (Algorithm 2).

The *Attention Module* performs similarity measurement between keys and queries which more details are provided in Algorithm 3 and Reference [20].

Algorithm 3 THE ATTENTION MODULE

function: Attention(input, key size d_k , value size d_v):

- 1: K = affine(input, d_k)
- 2: Q = affine(input, d_k)
- 3: V = affine(input, d_v)
- 4: logits = matmul(Q , transpose(K))
- 5: probs = softmax($\frac{\text{logits}}{\sqrt{d_k}}$)
- 6: output = matmul(probs, V)

return concat(input, output)

3. EXPERIMENTS AND RESULTS

In this section, the database used to evaluate the proposed architecture is first described. Then, the various experiments

will be discussed in the following sections. It is noteworthy to mention that we evaluated the proposed model for two scenarios. In all the experiments, for training purpose, the Adam optimizer with learning rate of 0.0001 was used. The loss function is obtained using Cross-entropy between the predicted and the truth label of final example in each $\mathcal{T}_j$. The mini-batch with size 64 also was used in the all experiments.

3.1. Database

In this paper, the 5th Ninapro database [21] referred to as the DB5 is used. DB5 dataset is recorded with two Thalmic Myo armbands including 16 active singledifferential wireless electrodes, recording muscular activity at a rate of 200 Hz. More specifically, the DB5 dataset consists of signals collected from 10 intact-limb users performing 52 movements including basic movements of the fingers, isometric, isotonic hand configurations, basic wrist movements, and grasping and functional movements. Each movement in the DB5 dataset is repeated 6 times, each lasting for 5 seconds followed by 3 seconds of rest. The DB5 dataset was presented in three sets of exercises which more details are provided in [21]. It is noteworthy to say that to follow the same criteria in [14, 15] and also have a fair comparison, in this paper, we only consider the lower armband in DB5.

3.2. Experiment 1

In this experiment, we evaluate the performance of the proposed TFS-HGR architecture for *new repetitions* with few-shot examples for DB5 dataset. We considered 2/3 of the gesture repetitions of each subject for training and validation purpose. The remaining repetitions constitute the test set. Table 1 shows the performance of model when FSL is applied for HGR task. In this experiment, we used two Embedding Modules, i.e., FC and LSTM Embedding. It can be observed that the Embedding Module could affect the overall accuracy. Moreover, by increasing the number of shots, the accuracy will be improved. It is noteworthy to mention that the performance of the proposed TFS-HGR architecture for this experiment is evaluated based on the second set of exercises DB5, which includes 17 movements.

3.3. Experiment 2

In this experiment, it is shown that the proposed TFS-HGR model can be used for new movements based on only a few

Table 2: Experiment 2: 5-way 1-shot, 5-way 5-shot, and 5-way 10-shot classification accuracies based on *new gestures with few-shot observation*. In this experiment, we adopted two different Embedding Modules: (i) FC Embedding and (ii) LSTM Embedding

The Embedding Module	5-way Accuracy (%) $\pm$ STD		
	1-shot	5-shot	10-shot
FC Embedding	40.23 $\pm$ 0.14	65.95 $\pm$ 0.15	75.29 $\pm$ 0.15
LSTM Embedding	39.56 $\pm$ 0.15	65.14 $\pm$ 0.14	74.91 $\pm$ 0.16

examples. More precisely, the test set consists of completely new gestures. This is a challenging task because unlike the Experiment 1, here, the probability distribution of the test set is not the same as training set. In other words, the model leverage the experience from source gestures to adapt to the variations in the new gestures in the target. Table 2 shows the efficiency of the proposed model when we had out-of-sample movements in the test set. The model predicted unknown class distributions in scenarios where few observations from the target distribution were available. It is worth to mention that in this experiment, 52 movements in DB5 are used. More specifically, 68% of total gestures are considered for the training purpose, while the validation set consists of 12% of the total gestures. The test set consists of the remaining gestures (20% of the total gestures).

4. CONCLUSION

In this work, a novel FSL architecture is proposed for HGR through sEMG signals. The proposed TFS-HGR architecture can generalize after seeing very few examples of each class, and then use the obtained knowledge for predicting the new classes. The ability to use the obtained knowledge during the training and adapt to the new classes is a feature of the proposed architectures. In this paper, we showed that by taking advantage of FSL, the proposed architecture can predict the new gestures in the target domain, and the learning does not have to start from the beginning. This paper is a huge step toward proposing and developing FSL framework for a target domain when there is a distribution mismatch with the source domain, e.g., new users or electrode shift. In this regard, the focus of the future works is developing more robust domain adapting algorithms for HGR tasks.

5. REFERENCES

- [1] C. Castellini, *et al.*, "Peripheral Machine Interfaces: Going Beyond Traditional Surface Electromyography," *Frontiers in Neurorobotics*, vol. 8, p. 22, 2014.
- [2] D. Farina, *et al.*, "The Extraction of Neural Information from the Surface EMG for the Control of Upper-limb Prostheses: Emerging Avenues and Challenges," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 22, no.4, pp. 797-809, 2014.
- [3] K. Lai and S. Yanushkevich, "An Ensemble of Knowledge Sharing Models for Dynamic Hand Gesture Recognition," *Int. Joint Conference on Neural Networks (IJCNN)*, 2020, pp. 1-7.
- [4] K. Lai and S. N. Yanushkevich, "CNN+RNN Depth and Skeleton based Dynamic Hand Gesture Recognition," *International Conference on Pattern Recognition (ICPR)*, 2018, pp. 3451-3456.
- [5] E. Rahimian, S. Zabihi, S. F. Atashzar, A. Asif, and A. Mohammadi, "Surface EMG-Based Hand Gesture Recognition via Hybrid and Dilated Deep Neural Network Architectures for Neurorobotic Prostheses," *Journal of Medical Robotics Research*, pp. 1-12, 2020.
- [6] E. Rahimian, S. Zabihi, F. Atashzar, A. Asif, A. Mohammadi, "XceptionTime: Independent Time-Window XceptionTime Architecture for Hand Gesture Classification," *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2020.
- [7] E. Rahimian, S. Zabihi, S. F. Atashzar, A. Asif, A. Mohammadi, "sEMG-Based Hand Gesture Recognition via Dilated Convolutional Neural Networks," *IEEE Global Conference on Signal and Information Processing (GlobalSIP)*, 2019.
- [8] E. Rahimian, S. Zabihi, A. Asif, D. Farina, S.F. Atashzar, and A. Mohammadi "FS-HGR: Few-shot Learning for Hand Gesture Recognition via ElectroMyography," *IEEE Trans. Neural Syst. Rehabil. Eng.*, 2021. In Press.
- [9] E. Rahimian, S. Zabihi, F. Atashzar, A. Asif, A. Mohammadi, "Few-Shot Learning for Decoding Surface Electromyography for Hand Gesture Recognition," *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2021, pp. 1300-1304.
- [10] W. Wei, *et al.*, "Surface Electromyography-based Gesture Recognition by Multi-view Deep Learning," *IEEE Trans. Biomedical Eng.*, vol. 66, no. 10, pp. 2964-2973, 2019.
- [11] Y. Hu, *et al.*, "A Novel Attention-based Hybrid CNN-RNN Architecture for sEMG-based Gesture Recognition," *PloS one*, vol. 13, no. 10, p.e0206049, 2018.
- [12] L. Chen, J. Fu, Y. Wu, H. Li, and B. Zheng, "Hand Gesture Recognition Using Compact CNN Via Surface Electromyography Signals," *Sensors*, vol. 20, no.3, p. 672, 2020.
- [13] W. Wei, *et al.* "A Multi-stream Convolutional Neural Network for sEMG-based Gesture Recognition in Muscle-computer Interface," *Pattern Recognition Letters*, 119, pp. 131-138, 2019.
- [14] U. Ct-Allard, *et al.* "Deep Learning for Electromyographic Hand Gesture Signal Classification using Transfer Learning," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 27, no. 4, pp. 760-771, 2019.
- [15] U. Ct-Allard, *et al.*, "Transfer learning for SEMG Hand Gestures Recognition Using Convolutional Neural Networks," *Proc. IEEE Int. Conf. Syst., Man, Cybern.*, Oct. 2017, pp. 16631668.
- [16] X. Zhang, L. Wu, B. Yu, X. Chen, and X. Chen, "Adaptive Calibration of Electrode Array Shifts Enables Robust Myoelectric Control," *IEEE Trans. Biomedical Eng.*, 2019.
- [17] M. Zia ur Rehman, *et al.*, "Multiday EMG-based Classification of Hand Motions with Deep Learning Techniques," *Sensors*, vol. 18, no. 8, p. 2497, 2018.

- [18] Y. Du, W. Jin, W. Wei, Y. Hu, and W. Geng, "Surface Emg-based Inter-session Gesture Recognition Enhanced by Deep Domain Adaptation," *Sensors*, vol. 17, no. 3, p. 458, 2017.
- [19] N. Mishra, M. Rohaninejad, X. Chen, and P. Abbeel, "A Simple Neural Attentive Meta-Learner," *arXiv preprint arXiv:1707.03141*, 2017.
- [20] A. Vaswani, N. Shazeer, J. Uszkoreit, L. Jones, A. Gomez N., L. Kaiser, and I. Polosukhin, "Attention is All You Need," *arXiv preprint arXiv:1706.03762*, 2017a.
- [21] S. Pizzolato, L. Tagliapietra, M. Cognolato, M. Reggiani, H. Muller, and M. Atzori, "Comparison of Six Electromyography Acquisition Setups on Hand Movement Classification Tasks," *PLoS ONE*, vol. 12, no. 10, pp. 1-7, 2017.